

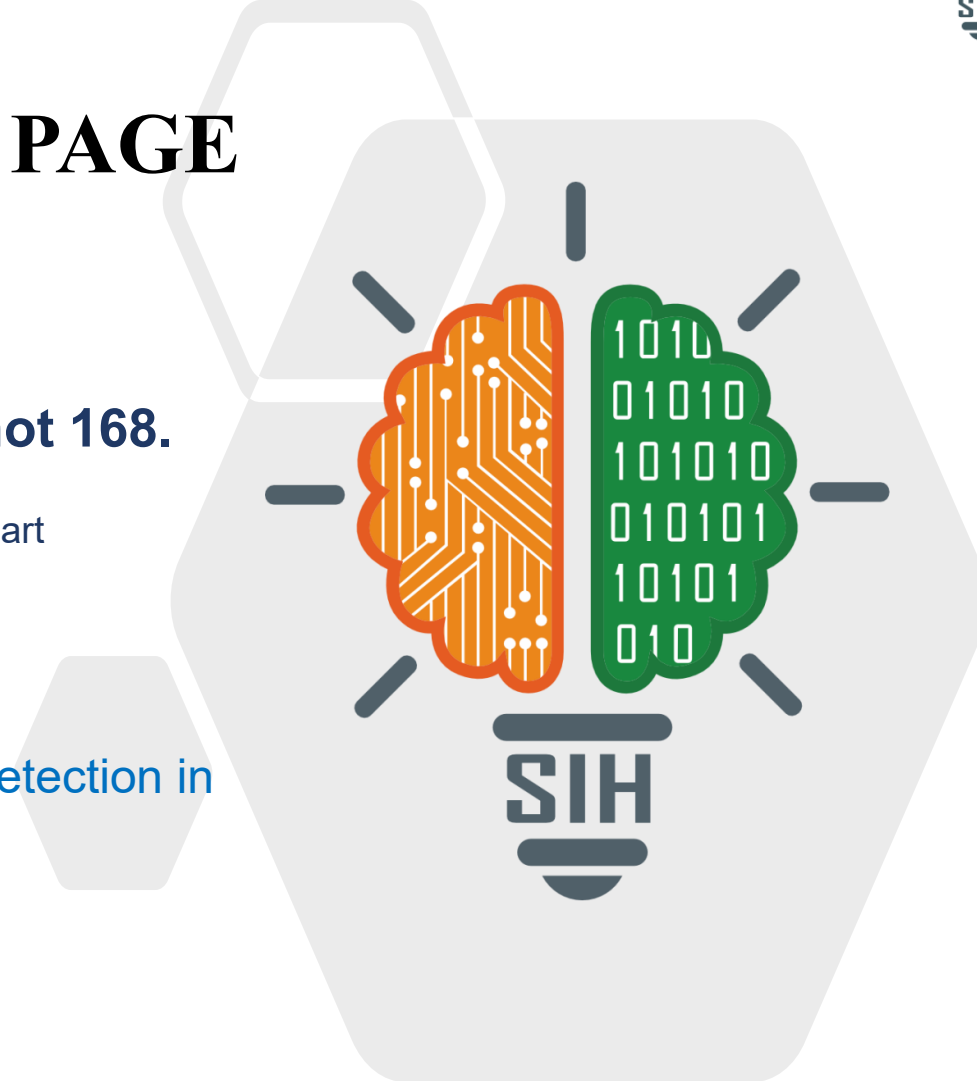
TITLE PAGE

AgniDrishti

Catch the drifting in-spec part at hour 24, not 168.

Decision support for ISRO burn-in QA: at the 24 h reading, each part is compared with its own lot and a 168 h forecast. Every decision shows its numbers and the rule that fired.

- **Problem Statement ID** – 26170
- **Problem Statement Title**- AI-Driven Anomaly Detection in Component Burn-In & Screening
- **Theme**- Smart Automation
- **PS Category**- Software
- **Team ID**- (to be assigned on the SIH portal)
- **Team Name**- Regnum Carya



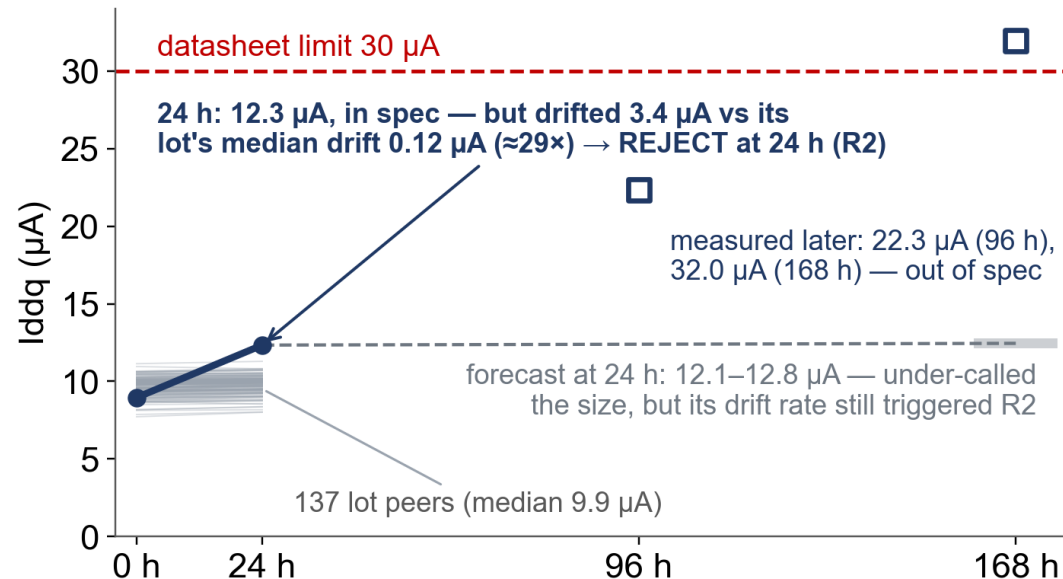
In spec is not the same as healthy

Proposed Solution (Describe your Idea/Solution/Prototype)

Detailed explanation of the proposed solution

Latent defects stay inside the datasheet limit but drift. **At hour 24, AgniDrishti compares each part with its own lot and a 168 h forecast.** Every decision, PASS included, shows its numbers and the rule that fired; bad or missing data is never PASSEd.

PIPELINE OUTPUT · synthetic held-out lot (not ISRO data) · datasheet check: PASS



How it addresses the problem

Module A level and 0→24 h drift vs the part's own lot (robust z)

Module B V0h + V24h → V168h forecast learned from past lots, with a 90 % range; early rejection if the drift rate exceeds the safety slope

Decision ordered rules R0–R6; the rule that fired is the explanation; doubtful parts go to an engineer (REVIEW)

Innovation and uniqueness of the solution

Drifting part-parameters caught at hour 24, of 30 (held-out synthetic lots; part-parameter = one part $\times$ one parameter)

4 / 30

datasheet limit at 24 h

4 / 30

+ DPAT lot level (AEC-Q001)

20 / 30

+ lot-relative drift

24 / 30

full AgniDrishti · 11 % flagged

Every layer is tested against a simpler baseline on held-out lots — the results include what did not help.

PS metric 1 · missed defects (test lots)
recall 0.80 (95 % CI 0.63–0.91); 11 % flagged

PS metric 2 · 168 h MAE (validation lots)
Iddq 0.297 vs 0.338 μA · tpd 0.047 vs 0.050 ns*

PS metric 3 · explainability
each decision shows its numbers + the rule fired

Technologies to be used (e.g. programming languages, frameworks, hardware)

Module B: 4 forecasters compared on validation MAE; a learned population increment (V24 + median 24→168 h rise) beat least-squares regression; 90 % split-conformal range.

Module A: robust lot statistics · **Decisions:** ordered rules, every output explainable · Python, web UI + CLI, offline, CPU only.

Methodology and process for implementation (Flow Charts/Images/ working prototype)



Decision rules — first match wins

- R0** bad or missing data
- R1** outside the datasheet limit now
- R2** drift rate > safety slope at optimistic end (Module B)
- R3** point or pessimistic end reaches slope or limit (B)
- R4** level and drift both ≥ 6 robust σ (Module A, DPAT)
- R5** level or drift ≥ 2.5 robust σ (Module A)
- R6** none of the above

- REVIEW
- REJECT
- REJECT
- REVIEW
- REJECT
- REVIEW
- PASS

robust z = $(x - \text{median}) / (\text{IQR} / 1.35) \cdot 6 \sigma$ DPAT, 2.5 σ tuned on validation
safety slope $s = \Delta_{\text{allow}} / 168 \text{ h}$, $\Delta_{\text{allow}} = 15 \% V_0$ · drift $r = (\hat{Y}_{168} - V_0) / 168 \text{ h}$

* vs persistence (predict V168 = V24). Flagged = REVIEW 10.4 % + REJECT 0.6 %.

Measured: held-out test lots, SYNTHETIC, evaluated once

Cumulative layers	Recall	Precision	REVIEW %	False rej.
E1 datasheet limit	0.133	0.125	1.2 %	0.0 %
E2 + DPAT lot level	0.133	0.085	1.9 %	0.0 %
E3 + lot-relative drift	0.667	0.213	3.5 %	0.3 %
E4 + 168 h forecast	0.667	0.213	3.3 %	0.3 %
E5 + 90 % range	0.667	0.213	3.4 %	0.3 %
E6 full ordered rules	0.800	0.092	10.4 %	0.3 %

Where the gain comes from: lot drift (4 → 20 of 30); ordered rules add +4 but raise REVIEW 3.4 % → 10.4 %. Precision falls as doubt goes to REVIEW, yet the flagged queue is $\approx 7 \times$ richer in drifting parts than the lot. The forecast adds no detection alone; it feeds R2/R3.

Potential challenges and risks	Strategies for overcoming these challenges
No official dataset for PS 26170	Synthetic PS-shaped lots, always labelled; site-fit script recalibrates on ISRO history (built; untested on real ISRO data)
PS names three metrics but no formula	Report all three: recall at a stated review share, 168 h MAE vs persistence, a rule-based explanation per part
“You detect the defects you injected”	Two extra behaviour families written outside the repo, sealed by SHA-256 on 26 Sep 2026, not yet opened — the final unseen test. Now: test lots scored once; leakage tests
Whole lot degrades / contaminated lot	Median / IQR statistics, plus absolute-limit and safety-slope backstops
Safety slope not defined in the PS	One Δ allow per site config (15 % of V0 in the prototype); per-parameter ESCC detail-spec Δ limits are the next step
Late-onset defects look healthy at 24 h (6 of 30 test escapes)	Stated limit; re-screen when the 96 h reading arrives (next)
Single glitchy 24 h reading → false REJECT: 8 of 14 REJECTs correct (6 false = 0.3 % of part-parameters)	REJECT is advisory: the engineer confirms (human-in-the-loop). v2: single-jump cases → REVIEW, tested on fresh pre-registered data

Analysis of the feasibility of the idea

What the QA engineer sees · prototype, slide 2's part

REJECT SYN-L026-C0010 · lot SYN-L026 · iddq → **REJECT**
(first rule R2)

- Data category: SYNTHETIC
- Measured: 0 h = 8.955 μ A; 24 h = 12.34 μ A; limits [n/a, 30] μ A → within the datasheet limit
- Lot position at 24 h: 4.63 robust-z from the lot median 9.907 μ A (robust scale 0.5247 μ A)
- Drift 0→24 h: 3.382 μ A, 42.7 robust-z from the lot's median drift 0.1159 μ A
- Predicted 168 h: 12.45 μ A; predicted drift rate 0.0208 μ A/h vs safety slope 0.008 μ A/h
- 90% prediction interval (target marginal coverage, not a failure probability): [12.14, 12.76] μ A

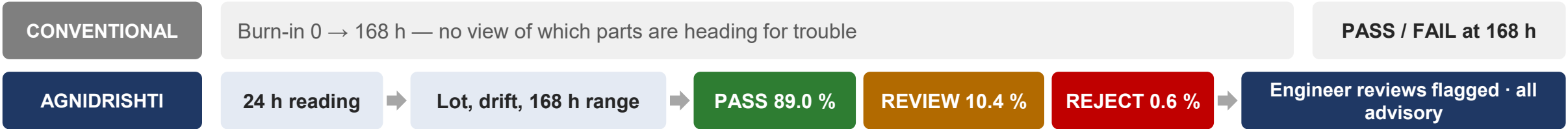
Scale 2,368 → 118,400 part-parameters: 0.35 s → 14.2 s, linear (one PC; test lots replicated 50×).

Built and verified · offline, CPU only
 ✓ 8 stages ✓ web UI + CLI ✓ any column layout (site config)
 ✓ 111 tests ✓ audit record + replay → identical decisions
github.com/Mrudula211/AgniDrishti/tree/p1-synthetic-data

Same burn-in. The engineer sees risk at hour 24, with the evidence.

Potential impact on the target audience

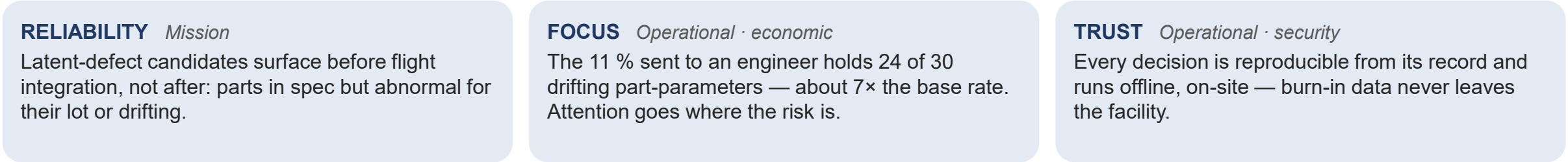
For: QA and reliability engineers screening space-grade electronics



Decision shares measured on the held-out SYNTHETIC test lots (2,368 part-parameters).

At the 24 h checkpoint the QA engineer gets a ranked REVIEW queue — each part with its numbers, its lot, its forecast range and the rule that fired — and records a decision with a reason. Decision support, not automation.

Benefits of the solution (social, economic, environmental, etc.)



Focus figures: synthetic test lots. Possible chamber-time saving: a confirmed reject could exit 144 h early, if the facility can reload that slot — not measured.



Gains on ISRO hardware will be measured after validation on representative ISRO data.

Details / Links of the reference and research work

OFFICIAL

SIH 2026 PS 26170 — AI-Driven Anomaly Detection in Component Burn-In & Screening (ISRO). sih.gov.in/sih2026PS

RELIABILITY PRACTICE (informed the design — no compliance claimed)

AEC-Q001 Rev-D, Guidelines for Part Average Testing, AEC, 2011 — our Module A baseline (DPAT)

ESCC Generic Specification No. 9000, Issue 10, ESA, 2018 — burn-in drift limits

ESCC Detail Specification No. 9202/045, Issue 6, ESA, 2020 — example per-parameter drift limits (IDD ± 75 nA; IOL ± 15 %)

Planned: methodology check on NASA PCoE MOSFET/IGBT ageing data (external, not ISRO)

What we add to this prior art: a decision at the 24 h checkpoint (not after burn-in) from lot-relative drift, a calibrated 168 h range, and a REVIEW route with per-part evidence.

PRIOR ART

Burn-in ML: I. Ahmed, P. Baraldi, E. Zio, H. Lewitschnig, Comput. Ind. Eng. 204, 111115, 2025 · doi:10.1016/j.cie.2025.111115

Lot health for burn-in: L. Langenberg, R. Pätzold, A. Khalid, “Process Data Analysis for Improved Burn-In Strategies Based on Complementary AI Models”, ESREL 2023

Burn-in outliers: US 8,010,310 B2, Method and apparatus for identifying outliers following burn-in testing, 2011

METHOD

Conformal prediction: J. Lei et al., JASA 113(523), 2018 · doi:10.1080/01621459.2017.1307116

Grouped (lot) data: R. Dunn, L. Wasserman, A. Ramdas, JASA 118(544), 2023 · doi:10.1080/01621459.2022.2060112

Degradation: C. J. Lu, W. O. Meeker, Technometrics 35(2), 1993 · doi:10.1080/00401706.1993.10485038

Reject option (REVIEW): C. K. Chow, IEEE Trans. Inf. Theory 16(1), 1970 · doi:10.1109/TIT.1970.1054406

OUR OWN EVIDENCE — CHECK IT

Code: github.com/Mrudula211/AgniDrishti/tree/p1-synthetic-data · commit 1f5d29a · 111 tests · E1–E6 frozen before test lots were scored once

Final unseen test: two behaviour families sealed 26 Sep 2026 (SHA-256 1a7de276...), not yet opened.